Feature	DDR3	DDR4	DDR5
Launch Year	2007	2014	2020
Standard Speed Range	$800 \text{ to } 2133 \text{ MT/s}$	$1600 \text{ to } 3200 \text{ MT/s}$ *(up to 5000+ overclocked)*	$4800 \text{ to } 8800+ \text{ MT/s}$
Operating Voltage	1.5V *(1.35V for DDR3L)*	1.2V	1.1V
Power Management	Controlled entirely by the motherboard	Controlled entirely by the motherboard	Moved onto the stick via an onboard **PMIC** (Power Management IC)
Channel Architecture	Single 64-bit channel per stick	Single 64-bit channel per stick	Dual independent 32-bit subchannels per stick
Max Consumer Stick Size	Up to 16 GB	Up to 64 GB	Up to $128\text{ GB}+$
Error Correction (ECC)	Only on specific enterprise-grade ECC modules	Only on specific enterprise-grade ECC modules	**On-die ECC** standard on all consumer sticks (improves stability)
Physical Pins	240 pins (Desktop) / 204 pins (Laptop)	288 pins (Desktop) / 260 pins (Laptop)	288 pins (Desktop) / 260 pins (Laptop)
Typical Use Cases	Legacy systems, old budget server nodes, vintage	Mainstream everyday PCs, mid-tier gaming builds,	High-end gaming rigs, 4K/8K video editing, heavy 3D rendering,

Feature	DDR3	DDR4	DDR5
	gaming rigs (e.g., Intel 3rd/4th Gen, AMD FX).	standard office laptops, and balanced budget workstations.	AI/Data processing workloads, and modern server deployments.